Table 1: Summary of biological, fishery, survey, and observation inputs used in the operating model (OM). Values were derived from or informed by the BSB assessment unless otherwise specified.

Description	**Value / Type**
Biological Information	
Mean recruitment (northern population)	16,000
Mean recruitment (southern population)	21,000
Standard deviation of recruitment random effects	0.75
Standard deviation of numbers-at-age 2+ random effects	0.20
Maximum age	8
Natural mortality (M)	0.4 for all ages
Fraction of the year when spawning occurs	0.5
Fishery Information	
Historical conditioning period	1989–2023
Selectivity for northern commercial fleet	Age-specific, ages 1–8+: 0.031, 0.421, 0.858, 1, 1, 1, 1, 1
Selectivity for northern recreational fleet	Age-specific, ages 1–8+: 0.036, 0.355, 0.573, 0.782, 0.868, 0.928, 1, 1
Selectivity for southern commercial fleet	Logistic; $a_{50} = 2.381$, $k = 0.402$ ($k = 1/\text{slope}$)
Selectivity for southern recreational fleet	Logistic; $a_{50} = 2.788$, $k = 0.849$ ($k = 1/\text{slope}$)
Catch observation CV for commercial fleet	0.05
Catch observation CV for recreational fleet	0.15
Fleet age-composition sampling distribution	Multinomial
ESS for fleet age composition	50
Survey Information	
Historical conditioning period	1989–2023
Selectivity for region 1 survey index	Age-specific, ages 1–8+: 0.091, 0.463, 0.905, 0.962, 1, 1, 1, 1
Selectivity for region 2 survey index	Age-specific, ages 1–8+: 0.362, 1, 1, 1, 1, 1, 1, 1
Survey-index observation CV	0.4
Survey age-composition sampling distribution	Multinomial
ESS for survey age composition	50
Catchability for region 1 survey index	7.370412×10^{-5}
Catchability for region 2 survey index	2.832577×10^{-5}

Table 2: Summary of operating model (OM) scenarios. Movement rates (μ) indicate seasonal probabilities of movement between regions. Bold rows indicate scenarios presented in the main text.

Population structure	Scenario	Movement direction	Level	Movement pattern	Recruitment	Movement for Pop 1	Movement for Pop 2
Natal homing	**NBL**	**Bidirectional**	**Low**	**Constant**	**IID lognormal deviations**	$\mu_{r1\to r2}$ **(Aug–Dec) = 0.022**	$\mu_{r2\to r1}$ **(Jan–Apr) = 0.313**
	NBH	Bidirectional	High	Constant	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.05	$\mu_{r2\to r1}$ (Jan–Apr) = 0.313
	NUL	Unidirectional	Low	Constant	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.022	$\mu_{r1\to r2}$ (Jan–Apr) = 0.022
	NUH	Unidirectional	High	Constant	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.05	$\mu_{r1\to r2}$ (Jan–Apr) = 0.05
	NBLI	Bidirectional	Low	IID by year	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.022	$\mu_{r2\to r1}$ (Jan–Apr) = 0.313
	NBLO	Bidirectional	Low	Ontogenetic	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.022	$\mu_{r2\to r1}$ (Jan–Apr) = 0.313
	NBLD	**Bidirectional**	**Low**	**Constant**	**Declining recruitment**	$\mu_{r1\to r2}$ **(Aug–Dec) = 0.022**	$\mu_{r2\to r1}$ **(Jan–May) = 0.313**
	NULD	**Unidirectional**	**Low**	**Constant**	**Declining recruitment**	$\mu_{r1\to r2}$ **(Aug–Dec) = 0.022**	$\mu_{r1\to r2}$ **(Jan–Apr) = 0.022**
	NBH-BH	Bidirectional	High	Constant	Beverton–Holt	$\mu_{r1\to r2}$ (Aug–Dec) = 0.05	$\mu_{r2\to r1}$ (Jan–May) = 0.313
Metapopulation	MBL	Bidirectional	Low	Constant	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.022	$\mu_{r2\to r1}$ (Jan–May) = 0.313
	MBH	**Bidirectional**	**High**	**Constant**	**IID lognormal deviations**	$\mu_{r1\to r2}$ **(Aug–Dec) = 0.05**	$\mu_{r2\to r1}$ **(Jan–May) = 0.313**
	MUL	**Unidirectional**	**Low**	**Constant**	**IID lognormal deviations**	$\mu_{r1\to r2}$ **(Aug–Dec) = 0.022**	$\mu_{r1\to r2}$ **(Jan–May) = 0.022**
	MUH	**Unidirectional**	**High**	**Constant**	**IID lognormal deviations**	$\mu_{r1\to r2}$ **(Aug–Dec) = 0.05**	$\mu_{r1\to r2}$ **(Jan–May) = 0.05**
	MBLI	Bidirectional	Low	IID by year	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.022	$\mu_{r2\to r1}$ (Jan–May) = 0.313
	MBLO	Bidirectional	Low	Ontogenetic	IID lognormal deviations	$\mu_{r1\to r2}$ (Aug–Dec) = 0.022	$\mu_{r2\to r1}$ (Jan–May) = 0.313
	MBH-BH	Bidirectional	High	Constant	Beverton–Holt	$\mu_{r1\to r2}$ (Aug–Dec) = 0.05	$\mu_{r2\to r1}$ (Jan–May) = 0.313

Table 3: Summary of catch allocation strategies used to distribute total catch advice from a single-region estimation model (EM) among regions and fleets. Strategies differ in the degree to which they incorporate recent fishing patterns and spatial signals of abundance. For strategies using empirical catch or survey information, allocation weights were calculated from recent three-year mean catch or survey-index values.

Strategy	**Allocation basis**	**Regional signal**	**Fleet signal**	**Equation**	**Description**
Equal	None	No	No	$C_{r,f} = \frac{C_{\text{total}}}{4}$	Total catch divided equally among all four region-fleet combinations, independent of spatial structure or fleet type.
Gear-Equal	Historical catch	No	Yes	$C_{r,fg} = C_{\text{total}} \times \frac{\sum_{r'} C^{\text{hist}}_{r',g}}{\sum_{r',g'} C^{\text{hist}}_{r',g'}} \times \frac{1}{2}$	Catch first allocated between commercial and recreational fleets according to historical total catches across regions, and then divided equally between regions within each fleet.
Fleet-Catch	Historical catch	Yes	Yes	$C_{r,f} = C_{\text{total}} \times \frac{C^{\text{hist}}_{r,f}}{\sum_{r',f'} C^{\text{hist}}_{r',f'}}$	Catch allocated in proportion to each region-fleet combination's historical share of total catch.
Index-Equal	Survey index	Yes	No	$C_{r,f} = C_{\text{total}} \times \frac{I^{\text{hist}}_{r}}{\sum_{r'} I^{\text{hist}}_{r'}} \times \frac{1}{2}$	Catch allocated among regions in proportion to historical survey indices, and then divided equally between commercial and recreational fleets within each region.
Index-Gear	Survey index + catch	Yes	Yes	$C_{r,fg} = C_{\text{total}} \times \frac{I^{\text{hist}}_{r}}{\sum_{r'} I^{\text{hist}}_{r'}} \times \frac{C^{\text{hist}}_{r,g}}{\sum_{g'} C^{\text{hist}}_{r,g'}}$	Catch allocated among regions in proportion to historical survey indices and, within each region, allocated between fleets in proportion to historical catches.

Note: r indexes regions ($r = 1, 2$); f indexes fleets (commercial and recreational fleets within each region); g indexes fleet type; the term gear in strategy names refers to commercial and recreational fleet types; C_{total} is total catch advice from the estimation model; $C^{\text{hist}}_{r,f}$ is historical mean catch for region r and fleet f; I^{hist}_{r} is the historical mean survey index for region r.

Table 4: Summary of performance metrics. Metrics are computed within each simulation realization and then averaged across N retained realizations. Catch and SSB are summarized over short-term and long-term windows, corresponding to the first k years and last ℓ years of the feedback period, respectively (here $k = \ell = 10$). Average annual variation and risk probabilities are summarized across the full feedback period. Metric scores are min–max scaled across catch allocation strategies within each operating model (OM) scenario and spatial scale, with variability and risk metrics inverted so that higher scores indicate better performance.

Performance metric	**Equation**	**Description**
Short-term average catch	$\tilde{x}^{(t)}_{o,r,\text{Catch_ST},m} = \frac{1}{k}\sum_{y=1}^{k} C^{(t)}_{o,r,y,m}$	Realization-level mean catch over the first k years of the feedback period.
Short-term average SSB	$\tilde{x}^{(t)}_{o,r,\text{SSB_ST},m} = \frac{1}{k}\sum_{y=1}^{k} \text{SSB}^{(t)}_{o,r,y,m}$	Realization-level mean SSB over the first k years of the feedback period.
Long-term average catch	$\tilde{x}^{(t)}_{o,r,\text{Catch_LT},m} = \frac{1}{\ell}\sum_{y=n-\ell+1}^{n} C^{(t)}_{o,r,y,m}$	Realization-level mean catch over the last ℓ years of the feedback period.
Long-term average SSB	$\tilde{x}^{(t)}_{o,r,\text{SSB_LT},m} = \frac{1}{\ell}\sum_{y=n-\ell+1}^{n} \text{SSB}^{(t)}_{o,r,y,m}$	Realization-level mean SSB over the last ℓ years of the feedback period.
Probability of overfishing $P[\text{OF}]$	$\tilde{x}^{(t)}_{o,r,\text{POF},m} = \frac{1}{n}\sum_{y=1}^{n} \mathbb{I}\left(\frac{F^{(t)}_{o,r,y,m}}{F^{(t)}_{o,r,40\%,y,m}} > 1\right)$	Realization-level proportion of feedback years with $F/F_{40\%} > 1$.
Probability of being overfished $P[\text{OFD}]$	$\tilde{x}^{(t)}_{o,r,\text{POFD},m} = \frac{1}{n}\sum_{y=1}^{n} \mathbb{I}\left(\frac{\text{SSB}^{(t)}_{o,r,y,m}}{\text{SSB}^{(t)}_{o,r,40\%,y,m}} < 0.5\right)$	Realization-level proportion of feedback years with $\text{SSB}/\text{SSB}_{40\%} < 0.5$.
Average annual catch variation	$\tilde{x}^{(t)}_{o,r,\text{AAVC},m} = \frac{\frac{1}{n-1}\sum_{y=2}^{n}\left\lvert C^{(t)}_{o,r,y,m} - C^{(t)}_{o,r,y-1,m}\right\rvert}{\frac{1}{n}\sum_{y=1}^{n} C^{(t)}_{o,r,y,m}}$	Realization-level mean absolute annual change in catch, scaled by mean catch across the feedback period.
Average annual SSB variation	$\tilde{x}^{(t)}_{o,r,\text{AAVS},m} = \frac{\frac{1}{n-1}\sum_{y=2}^{n}\left\lvert \text{SSB}^{(t)}_{o,r,y,m} - \text{SSB}^{(t)}_{o,r,y-1,m}\right\rvert}{\frac{1}{n}\sum_{y=1}^{n} \text{SSB}^{(t)}_{o,r,y,m}}$	Realization-level mean absolute annual change in SSB, scaled by mean SSB across the feedback period.
Aggregated metric value (across realizations)	$x_{o,r,j,m} = \frac{1}{N}\sum_{t=1}^{N} \tilde{x}^{(t)}_{o,r,j,m}$	Mean across N retained realizations of the realization-level metric values.
Metric score (0–1; "higher is better")	$s_{o,r,j,m} = \frac{x_{o,r,j,m} - \min_{m'} x_{o,r,j,m'}}{\max_{m'} x_{o,r,j,m'} - \min_{m'} x_{o,r,j,m'}}, \quad s_{o,r,j,m} \in [0,1]$	Min–max scaling for metric j across strategies m within each OM scenario o and spatial scale r, when larger values indicate better performance, such as catch or SSB.
Metric score (0–1; "lower is better")	$s_{o,r,j,m} = 1 - \frac{x_{o,r,j,m} - \min_{m'} x_{o,r,j,m'}}{\max_{m'} x_{o,r,j,m'} - \min_{m'} x_{o,r,j,m'}}, \quad s_{o,r,j,m} \in [0,1]$	Min–max scaling for metric j across strategies m within each OM scenario o and spatial scale r, when smaller values indicate better performance, such as variability or risk.
Average scaled performance score	$A_{o,r,m} = \frac{1}{J}\sum_{j=1}^{J} s_{o,r,j,m}, \qquad J = 8, \quad A_{o,r,m} \in [0,1]$	Equal-weight descriptive summary across the eight metric scores used in the multi-metric performance figures; values range from 0 to 1.

Note: o indexes OM scenarios; r indexes spatial scale (region 1, region 2, and the system-level aggregate); m indexes catch allocation strategies; t indexes simulation realizations ($t = 1, \ldots, N$); j indexes performance metrics; k and ℓ denote the short- and long-term windows; n is the number of years in the feedback period; $\mathbb{I}(\cdot)$ is an indicator function. Scores are computed separately for each (o, r, j) by scaling across strategies m.

Table 5: Summary of empirical allocation strategy tradeoffs, relative performance patterns in the MSE, and management situations for which each strategy may be most appropriate.

Strategy	**Information used**	**Main advantage**	**Main limitation**	**Best suited for**
Equal	No regional or fleet information	Transparent, stable, and low data demand.	Could not respond to directional movement or regional productivity decline, increasing localized depletion risk in vulnerable components.	Situations where regional productivity and depletion risk are expected to be similar, or where transparency and catch stability are primary objectives.
Gear-Equal	Historical fleet catch shares; equal regional split within each fleet	Produced stable catch outcomes while preserving broad fleet allocation patterns.	Lacked regional biological information, so regional risk could remain hidden when movement or productivity differed among regions.	Situations where maintaining broad fleet continuity is important and persistent regional differences in productivity or risk are not expected to dominate.
Fleet-Catch	Recent x-year mean region-fleet catch shares	Incorporated recent fishing patterns, fleet access, and spatial catch distribution; provided an intermediate level of variability.	Catch reflected regulations, profitability, access, reporting, and historical fishing patterns as well as biomass, so allocation could lag biological change or regional productivity shifts.	Situations where recent catch is expected to reasonably reflect resource availability, fleet access, and spatial fishing opportunities.
Index-Equal	Recent x-year mean regional survey indices; equal fleet split within each region	Responded directly to regional abundance signals and could reduce localized depletion risk when survey signals tracked regional risk.	Increased interannual catch variability; survey indices could reflect seasonal availability rather than annual exploitable biomass or spawning contribution.	Situations where regional survey indices reliably track relative abundance or depletion risk and managers can tolerate greater catch variability.
Index-Gear	Recent x-year mean regional survey indices plus fleet catch shares within each region	Combined a regional abundance signal with fleet allocation history.	Shared the survey-timing and variability concerns of Index-Equal while also depending on catch shares within regions.	Situations where regional survey responsiveness is desired, but fleet-specific catch history and access also need to be retained.

Note: x is the smoothing window used to calculate recent empirical catch or survey-index allocation weights; in this study, $x = 3$ years.

Table S1: Summary of movement and recruitment formulations used in operating model scenarios. Scenario abbreviations denote combinations of population structure and process representation: N = natal homing, M = metapopulation; B = bidirectional, U = unidirectional movement; L = low movement magnitude; H = high movement magnitude; I = IID interannual movement, O = ontogenetic movement; D = declining recruitment in population component 1; BH = density-dependent Beverton–Holt recruitment.

Scenario(s)	**Process**	**Equation**	**Description**
NBLI, MBLI	IID_y movement	$\text{logit}(m_{i,a,r\to r',y}) = \text{logit}(\bar{m}_{i,a,r\to r'}) + \delta_{i,r\to r',y}$, $\delta_{i,r\to r',y} \sim \mathcal{N}(0, \sigma^2_{\text{move}})$, $\sigma_{\text{move}} = 0.5$	Mean movement rates vary stochastically among years via IID random effects on the logit scale. Directionality constraints (bidirectional or unidirectional) are enforced by fixing prohibited transitions to zero.
NBLO, MBLO	Ontogenetic movement	$m_{i,a,r\to r'} = \dfrac{m^{\max}_{i,r\to r'}}{1+\exp[-k(a-a_{50})]}$	Movement probability increases systematically with age following a logistic function, representing ontogenetic dispersal behavior. Parameters are $k = 2$ (steepness) and $a_{50} = 1$ (age at which 50% of maximum movement is reached).
NBLD, NULD	Declining recruitment	$\log(R_{1,y}) = \log(\bar{R}_{1,0}) - d_R(y - y_0) + \varepsilon^R_{1,y}$, $\varepsilon^R_{1,y} \sim \mathcal{N}(0, \sigma^2_R)$, $d_R = 0.016$ yr^{-1}, $\sigma_R = 0.75$	Recruitment varies stochastically around a declining time-varying mean. For population component 1, mean recruitment declines at a fixed rate of 0.016 yr^{-1} in log space during the feedback period, representing a cumulative 0.8 reduction in log recruitment over 50 years. Population component 2 remains stationary around its mean recruitment.
NBH-BH, MBH-BH	Density-dependent recruitment	$\bar{R}_{i,y} = \dfrac{\alpha_i\, SSB_{i,y}}{1+\beta_i\, SSB_{i,y}}$, $\log(R_{i,y}) = \log(\bar{R}_{i,y}) + \varepsilon^R_{i,y}$, $\varepsilon^R_{i,y} \sim \mathcal{N}(0, \sigma^2_R)$, $\sigma_R = 0.75$	Mean recruitment follows a Beverton–Holt stock–recruitment relationship with lognormal deviations. Parameters are $(\alpha_1, \beta_1) = (4.5, 10^{-4})$ and $(\alpha_2, \beta_2) = (4.0, 10^{-4})$.

Table S2: Summary of model features estimated, fixed, or derived in the single-region estimation model (EM).

Model component	**Treatment in EM**	**Description**
Spatial structure	Fixed	Single-region; no explicit regions, movement, or population components.
Mean recruitment	Estimated	Aggregate mean.
Recruitment deviation SD	Estimated	IID lognormal recruitment deviations.
Numbers-at-age 2+ deviation SD	Estimated	IID lognormal age-2+ deviations.
Initial numbers-at-age	Equilibrium initialization	WHAM equilibrium approach.
Annual fishing mortality by fleet	Estimated	Commercial and recreational fleets.
Commercial selectivity	Partly estimated	Ages 1–3 estimated; ages 4–8 fixed at 1.
Recreational selectivity	Partly estimated	Ages 1–4 estimated; ages 5–8 fixed at 1.
Survey selectivity	Partly estimated	Age 1 estimated; ages 2–8 fixed at 1.
Survey catchability	Estimated	Aggregate survey index.
Catch observation error CVs	Fixed	Commercial = 0.05; recreational = 0.15.
Survey-index observation error CV	Fixed	CV = 0.4.
Age-composition ESS	Fixed	ESS = 50.
Natural mortality (M)	Fixed	0.4 for all ages.
Weight-at-age	Fixed inputs	Aggregated before fitting EM.
Maturity-at-age	Fixed inputs	Aggregated before fitting EM.
$F_{40\%}$	Derived	Calculated from each EM fit.